Aegis Ledger

A permissioned blockchain concept for national defense supply chains and access records

Concept Draft

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Overview

This concept describes Aegis Ledger: a permissioned blockchain used as an immutable national ledger. It records critical defense supply-chain events and system access logs. Authorized parties share one consistent history. No single organization controls the record.

Problem

Defense supply chains and sensitive systems currently rely on separate databases. A single compromised server can alter records, hide shipments, or erase access logs. Cross-agency visibility is slow and incomplete. Trust between organizations depends on bilateral agreements and manual audits.

Solution

A permissioned blockchain holds the ledger. Each critical event — a component leaving a factory, a clearance granted, a system login — is written once. Multiple independent nodes, operated by different agencies or trusted partners, must agree before the entry is final. Once written, the entry cannot be changed or deleted.

How it works

Participants run nodes. Write access is limited to authorized roles. Read access is granted by policy. The ledger stores only the minimum required data: identifiers, timestamps, status changes, and cryptographic hashes of supporting documents. Full documents stay in existing secure stores; the blockchain records that they existed and were not altered.

The system is designed as a layer on top of current processes. It does not replace existing systems. It adds a shared, unchangeable record of what those systems reported.

Properties

- No single point of failure or control.
- Full history visible to authorized parties in real time.
- Tamper-evident by design.
- Clear attribution of every change.
- Operates alongside existing infrastructure.

Status

This is a concept in design. No system has been built, tested, or deployed. The architecture described here is a proposal for further definition.

Private. Not for distribution without approval.